Evaluate each integral

1. $\int xe^{2x}dx$

2. $\int \frac{\ln(x)}{x^3}dx$

3. $\int x^2e^{3x}dx$

4. $\int x^2 \tan^{-1}(x)$

5. $\int e^{3x} \sin(2x) dx$

6. $\int e^{\sqrt{x}} dx.$

7. $\int x^3 e^{x^2} dx$

8. $\int \sin(\ln(x)) dx$